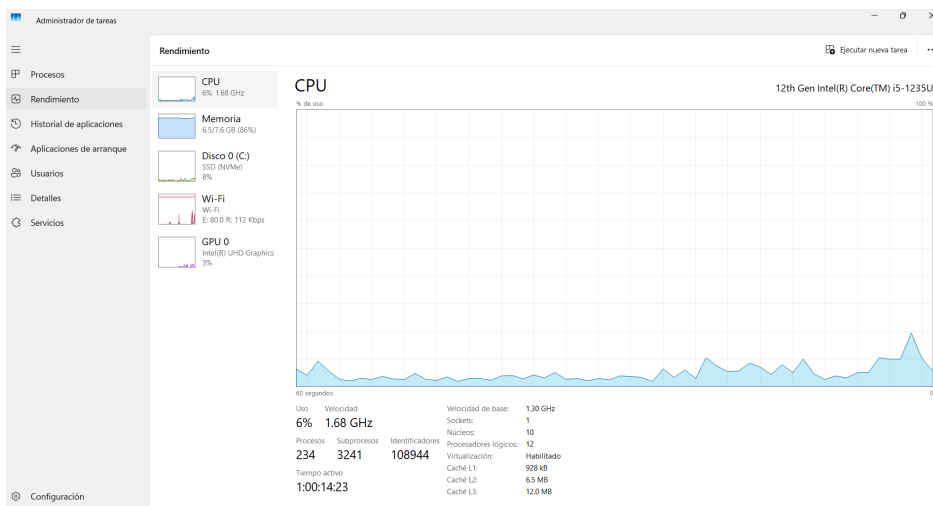


UNIVERSIDAD MARIANO GÁLVEZ
FACULTAD DE INGENIERÍA EN SISTEMAS DE INFORMACIÓN Y
CIENCIAS DE LA COMPUTACIÓN
INTRODUCCIÓN A LOS SISTEMAS DE CÓMPUTO



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Desktop Sharing

☒

Remote Control

☐

Allows desktop shares to control the screen

How to Connect

Use a remote desktop app to connect using the RDP protocol. Additional information about the device's network location may also be required.

Hostname

deydry

Port

3389

Login Details

Username

deyVB

Password

.....

```
vboxuser@deydry: ~  
vboxuser@deydry:~$ whoami  
vboxuser  
vboxuser@deydry:~$ hostname  
deydry  
vboxuser@deydry:~$ uname -a  
Linux deydry 6.17.0-14-generic #14~24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Thu Jan 15  
15:52:10 UTC 2 x86_64 x86_64 x86_64 GNU/Linux  
vboxuser@deydry:~$ lsb_release -a  
No LSB modules are available.  
Distributor ID: Ubuntu  
Description:    Ubuntu 24.04.4 LTS  
Release:        24.04  
Codename:       noble  
vboxuser@deydry:~$
```

¿Cuál es tu usuario? deyVB

• ¿Qué versión de Ubuntu instalaste

Versión: 7.2.6a

```
username: command not found  
vboxuser@deydry:~$ pwd  
/home/vboxuser  
vboxuser@deydry:~$ ls  
Desktop Documents Downloads Music Pictures Public snap Templates Videos  
vboxuser@deydry:~$ cd  
vboxuser@deydry:~$ cd ~  
vboxuser@deydry:~$ cd~  
Command 'cd~' not found, did you mean:  
  command 'cd5' from deb cd5 (0.1-4)  
  command 'cdo' from deb cdo (2.3.0-1)  
  command 'cdi' from deb cdo (2.3.0-1)  
  command 'cdp' from deb irpas (0.10-9)  
  command 'cdb' from deb tinycdb (0.81-1)  
  command 'cdw' from deb cdw (0.8.1-3)  
  command 'cde' from deb cde (0.1+git9-g551e54d-1.2)  
Try: sudo apt install <deb name>  
vboxuser@deydry:~$ mkdir semana3  
vboxuser@deydry:~$ cd semana3  
vboxuser@deydry:~/semana3$ touch notas.txt  
vboxuser@deydry:~/semana3$ echo "Hola UMG" > notas.txt  
vboxuser@deydry:~/semana3$ cat notas.txt  
Hola UMG  
vboxuser@deydry:~/semana3$
```

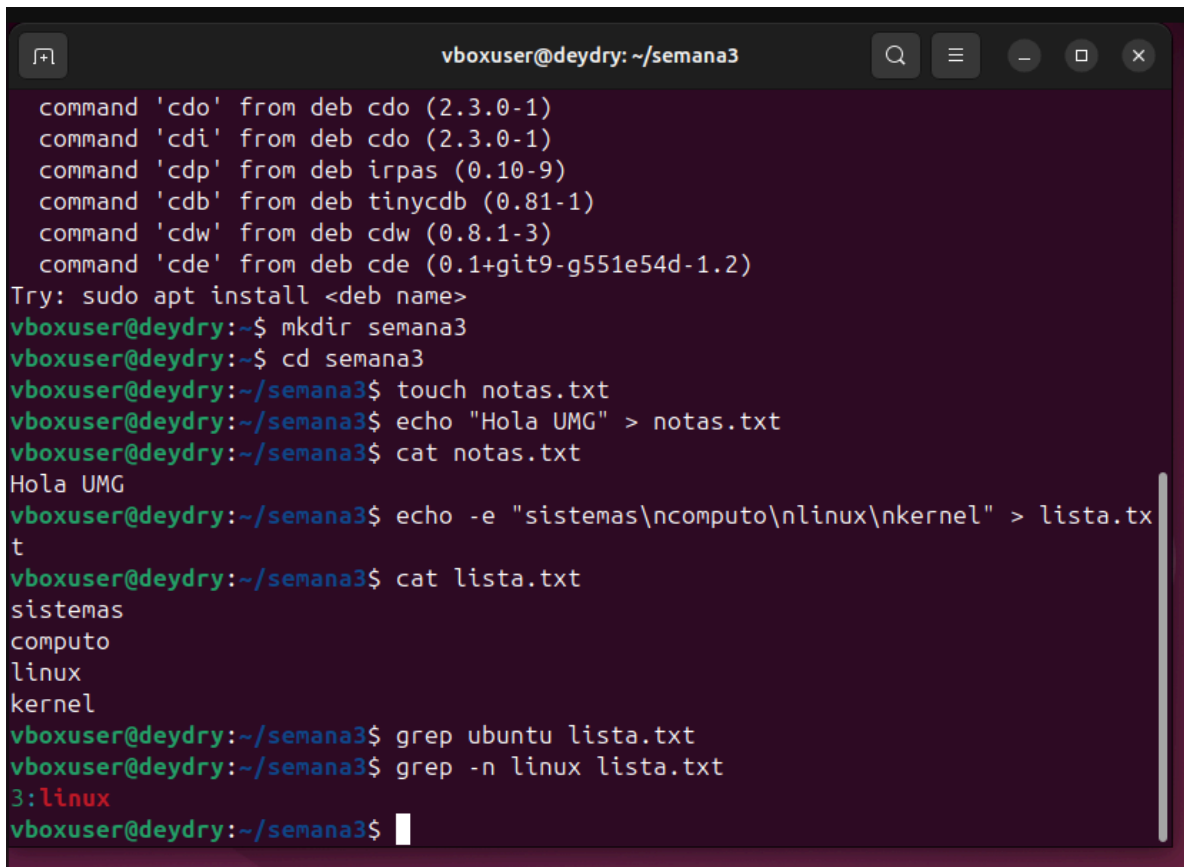
¿Qué hace pwd?

Muestra la ruta a la carpeta (Print Working directory)

¿Qué diferencia hay entre '>' y '>>'?

> Elimina el texto y lo reemplaza

>> añade a lo que ya estaba escrito

A terminal window titled 'vboxuser@deydry: ~/semana3' with standard window controls. The terminal shows a series of commands and their outputs. It starts with listing several 'command' entries from different Debian packages. Then, it prompts the user to run 'sudo apt install <deb name>'. The user then creates a directory 'semana3', changes to it, creates a file 'notas.txt', and writes 'Hola UMG' to it using the '>' redirection operator. Next, the user creates 'lista.txt' with the content 'sistemas\ncomputo\nlinux\nkernel' using the '>>' redirection operator. Finally, the user runs 'grep ubuntu lista.txt' (which returns nothing) and 'grep -n linux lista.txt', which returns '3:linux'.

Qué hace la opción -n en grep?

```
vboxuser@deydry:~/semana3$ ps aux | head
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root            1  0.1  0.3  23164 14552 ?        Ss   21:36   0:01 /sbin/init sp
lash
root            2  0.0  0.0      0     0 ?        S    21:36   0:00 [kthreadd]
root            3  0.0  0.0      0     0 ?        S    21:36   0:00 [pool_workque
ue_release]
root            4  0.0  0.0      0     0 ?        I<   21:36   0:00 [kworker/R-rc
u_gp]
root            5  0.0  0.0      0     0 ?        I<   21:36   0:00 [kworker/R-sy
nc_wq]
root            6  0.0  0.0      0     0 ?        I<   21:36   0:00 [kworker/R-kv
free_rcu_reclaim]
root            7  0.0  0.0      0     0 ?        I<   21:36   0:00 [kworker/R-sl
ub_flushwq]
root            8  0.0  0.0      0     0 ?        I<   21:36   0:00 [kworker/R-ne
tns]
root           11  0.0  0.0      0     0 ?        I<   21:36   0:00 [kworker/0:0H
-kblockd]
vboxuser@deydry:~/semana3$ top
```

```
vboxuser@deydry: ~/semana3
top - 22:02:04 up 25 min,  1 user,  load average: 0.07, 0.12, 0.10
Tasks: 195 total,  1 running, 194 sleeping,  0 stopped,  0 zombie
%Cpu(s):  1.9 us,  3.8 sy,  0.0 ni, 93.9 id,  0.0 wa,  0.0 hi,  0.3 si,  0.0 st
MiB Mem :  3914.8 total,  1713.4 free,  1080.6 used,  1369.4 buff/cache
MiB Swap:   0.0 total,   0.0 free,   0.0 used.  2834.2 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM     TIME+ COMMAND
 1868 vboxuser  20   0 3993192 377308 148140 S   14.3   9.4   0:48.87 gnome-s+
 2838 vboxuser  20   0 562964  54772  43732 S    3.2   1.4   0:03.69 gnome-t+
   69 root      20   0      0      0      0 I    1.6   0.0   0:02.78 kworker+
  138 root      20   0      0      0      0 I    0.3   0.0   0:02.76 kworker+
    1 root      20   0  23164  14552   9876 S    0.0   0.4   0:01.88 systemd
    2 root      20   0      0      0      0 S    0.0   0.0   0:00.01 kthreadd
    3 root      20   0      0      0      0 S    0.0   0.0   0:00.00 pool_wo+
    4 root       0 -20      0      0      0 I    0.0   0.0   0:00.00 kworker+
    5 root       0 -20      0      0      0 I    0.0   0.0   0:00.00 kworker+
    6 root       0 -20      0      0      0 I    0.0   0.0   0:00.00 kworker+
    7 root       0 -20      0      0      0 I    0.0   0.0   0:00.00 kworker+
    8 root       0 -20      0      0      0 I    0.0   0.0   0:00.00 kworker+
   11 root       0 -20      0      0      0 I    0.0   0.0   0:00.00 kworker+
   12 root      20   0      0      0      0 I    0.0   0.0   0:00.00 kworker+
   13 root       0 -20      0      0      0 I    0.0   0.0   0:00.00 kworker+
   14 root      20   0      0      0      0 S    0.0   0.0   0:00.21 ksoftir+
   15 root      20   0      0      0      0 I    0.0   0.0   0:00.74 rcu_pre+
```

En top, identifica el proceso con más CPU.

1868 vboxuser 14.3%

- ¿Cómo salís del top? (pista: tecla)

ESC

- ¿Qué diferencia hay entre ps y top?

PS- muestra una instantánea estática de los procesos en ejecución en el momento en que se ejecuta el comando.

TOP- Muestra una vista dinámica y en tiempo real de los procesos.

```
vboxuser@deydry:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          3.8Gi        1.1Gi        1.7Gi         32Mi        1.3Gi        2.8Gi
Swap:          0B           0B           0B

vboxuser@deydry:~$ df -h
df-h: command not found
vboxuser@deydry:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            392M  1.5M  391M   1% /run
/dev/sda2        25G   6.0G   18G  26% /
tmpfs            2.0G   0     2.0G   0% /dev/shm
tmpfs            5.0M   8.0K   5.0M   1% /run/lock
tmpfs            392M  132K  392M   1% /run/user/1000
vboxuser@deydry:~$ du -sh ~
16M    /home/vboxuser
vboxuser@deydry:~$
```

Cuánta RAM total detecta Ubuntu?

3.8 GB

- ¿Cuánto espacio libre queda en el disco?

2.8 GB

```
vboxuser@deydrv:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ec:b2:e9 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 84699sec preferred_lft 84699sec
    inet6 fd17:625c:f037:2:d182:fdc0:4d56:3d99/64 scope global temporary dynamic
        valid_lft 86339sec preferred_lft 14339sec
    inet6 fd17:625c:f037:2:a00:27ff:feec:b2e9/64 scope global dynamic mngtnpaddr
        valid_lft 86339sec preferred_lft 14339sec
    inet6 fe80::a00:27ff:feec:b2e9/64 scope link
        valid_lft forever preferred_lft forever
vboxuser@deydrv:~$
```

```
vboxuser@deydrv:~$ ping -c 4 google.com
PING google.com (192.178.50.46) 56(84) bytes of data.
64 bytes from lcmiaa-aa-in-f14.1e100.net (192.178.50.46): icmp_seq=1 ttl=255 time=37.1 ms
64 bytes from lcmiaa-aa-in-f14.1e100.net (192.178.50.46): icmp_seq=2 ttl=255 time=34.4 ms
64 bytes from lcmiaa-aa-in-f14.1e100.net (192.178.50.46): icmp_seq=3 ttl=255 time=36.0 ms
64 bytes from lcmiaa-aa-in-f14.1e100.net (192.178.50.46): icmp_seq=4 ttl=255 time=35.7 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3314ms
rtt min/avg/max/mdev = 34.350/35.775/37.083/0.974 ms
vboxuser@deydrv:~$
```



Network

Adapter 1: Intel PRO/1000 MT Desktop (NAT)

Responde:

- ¿Tu VM tiene IP asignada?

si

- ¿El ping respondió? (sí/no)

si

```

vboxuser@deydry:~$ ls -l
total 40
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Desktop
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Documents
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Downloads
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Public
drwxrwxr-x 2 vboxuser vboxuser 4096 Mar  4 21:58 semana3
drwx----- 3 vboxuser vboxuser 4096 Mar  4 17:35 snap
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Videos
vboxuser@deydry:~$ chmod 600 notas.txt
chmod: cannot access 'notas.txt': No such file or directory
vboxuser@deydry:~$ ls -l notas.txt
ls: cannot access 'notas.txt': No such file or directory
vboxuser@deydry:~$

```

```

vboxuser@deydry: ~
└─735 /usr/sbin/NetworkManager --no-daemon

Warning: some journal files were not opened due to insufficient permissions.
vboxuser@deydry:~$ chmod 600
chmod: missing operand after '600'
Try 'chmod --help' for more information.
vboxuser@deydry:~$ chmod 600 s
semana3/ snap/
vboxuser@deydry:~$ chmod 600 s
semana3/ snap/
vboxuser@deydry:~$ chmod 600 semana3/notas.txt
vboxuser@deydry:~$ ls -l
total 40
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Desktop
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Documents
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Downloads
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Public
drwxrwxr-x 2 vboxuser vboxuser 4096 Mar  4 21:58 semana3
drwx----- 4 vboxuser vboxuser 4096 Mar  5 00:00 snap
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Mar  4 17:35 Videos
vboxuser@deydry:~$

```

¿Qué cambió en los permisos después de chmod 600?

Se habilito un permiso mas.

```

Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
vboxuser@deydry:~$ neofetch

  .-/+00ssssso+/-.
  `:+ssssssssssssssss+:`
    -+ssssssssssssssssyyssss+-
      .ossssssssssssssssdMMMMyssso.
        /ssssssssshdmmNNmmymNMMMMhsssss/
          +ssssssshmydMMMMMMNddddyssssss+
            /ssssssshNMMMyhhyyyhmNMMMNhssssss/
              .sssssssdMMMNhssssssshNMMMdssssss.
                +ssssshhhyNMMNyssssssssssyNMMMyssssss+
                  ossyNMMMNyMMhssssssssssshmmhssssssso
                    ossyNMMMNyMMhssssssssssshmmhssssssso
                      +ssssshhhyNMMNyssssssssssyNMMMyssssss+
                        .sssssssdMMMNhssssssshNMMMdssssss.
                          /ssssssshNMMMyhhyyyhdNMMMNhssssss/
                            +sssssssdmydMMMMMMNddddyssssss+
                              /ssssssssshdmmNNNNmyNMMMNhssssss/
                                .ossssssssssssssssdMMMMyssso.
                                  -+ssssssssssssssssyyssss+-
                                    `:+ssssssssssssssss+:`
                                      .-/+00ssssso+/-.

vboxuser@deydry
-----
OS: Ubuntu 24.04.4 LTS x86_64
Host: VirtualBox 1.2
Kernel: 6.17.0-14-generic
Uptime: 33 mins
Packages: 1564 (dpkg), 10 (snap)
Shell: bash 5.2.21
Resolution: 1280x800
DE: GNOME 46.0
WM: Mutter
WM Theme: Adwaita
Theme: Yaru [GTK2/3]
Icons: Yaru [GTK2/3]
Terminal: gnome-terminal
CPU: 12th Gen Intel i5-1235U (2) @ 2
GPU: 00:02.0 VMware SVGA II Adapter
Memory: 894MiB / 3914MiB

```

¿Qué información te muestra neofetch?

muestra información del sistema de manera estética

```

vboxuser@deydry:~$ systemctl status NetworkManager --no-pager
Unit NetworkManager.service could not be found.
vboxuser@deydry:~$ systemctl status NetworkManager --no-pager
● NetworkManager.service - Network Manager
   Loaded: loaded (/usr/lib/systemd/system/NetworkManager.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-03-04 21:36:47 UTC; 35min ago
     Docs: man:NetworkManager(8)
    Main PID: 735 (NetworkManager)
      Tasks: 4 (limit: 4598)
     Memory: 19.1M (peak: 34.9M)
        CPU: 621ms
       CGroup: /system.slice/NetworkManager.service
               └─735 /usr/sbin/NetworkManager --no-daemon

Warning: some journal files were not opened due to insufficient permissions.
vboxuser@deydry:~$

```

¿El servicio está 'active (running)'?

si

8) Preguntas de reflexión

23. ¿Qué recurso se vuelve cuello de botella primero en una VM: CPU, RAM o Disco?
¿Por qué?

RAM será el primer cuello de botella en la mayoría de escenarios. El CPU de 2 núcleos puede manejar tareas básicas

24. ¿Qué ventaja tiene virtualizar Ubuntu en lugar de instalarlo directo en el disco del PC?

Portabilidad de las imágenes para poder montarlo en otro servidor. Poder tener varios sistemas operativos para tenerlo en un solo sistema físico.

25. ¿Qué te costó más: instalación, Guest Additions o comandos? ¿Por qué?

Comandos porque son procesos que no se que realmente que hacen asi que tuve que investigar y ver el resultado que me daba cada comando